50 Quiz Questions - Foundations of Computer Science

1. Let A be a language decided by a deterministic polynomial-time algorithm. What can we conclude?

A. A in NP and A in coNP

B. A not in NP but A in P

C. A in NP, but it is undecidable

D. A is not closed under reduction

Answer: A

2. Which of the following implications always holds?

A. A in NP -> A in P

B. A in coNP -> A in NP

C. A in P -> A in NP

D. A in NP -> not A in NP

Answer: C

3. A verifier for a language A runs in polynomial time and receives input x and certificate y. What must be true for A in NP?

A. The certificate must always be of constant size

B. For every x in A, there exists a certificate y of polynomial length such that the verifier accepts (x, y)

C. The verifier must reject all inputs x not in A for every certificate

D. The verifier must explore all possible certificates in polynomial time

Answer: B

4. Which of the following problems is in P?

A. Given a Boolean formula, is it satisfiable?

B. Given a graph and two vertices, is there a path between them?

C. Given a graph, does it have a clique of size k?

D. Given an integer n, is it the product of two primes?

Answer: B

5. If A <=p B and B in P, then:

A. A in NP

B. A in P

C. B is NP-complete

D. A = B

Answer: B

6. Let A be a language decided by a deterministic polynomial-time algorithm. What can we conclude?

A. A in NP and A in coNP

B. A not in NP but A in P

C. A in NP, but it is undecidable

D. A is not closed under reduction

Answer: A

7. Which of the following implications always holds?

A. A in NP -> A in P

B. A in coNP -> A in NP

C. A in P -> A in NP

D. A in NP -> not A in NP

Answer: C

8. A verifier for a language A runs in polynomial time and receives input x and certificate y. What must be true for A in NP?

A. The certificate must always be of constant size

B. For every x in A, there exists a certificate y of polynomial length such that the verifier accepts (x, y)

C. The verifier must reject all inputs x not in A for every certificate

D. The verifier must explore all possible certificates in polynomial time

Answer: B

9. Which of the following problems is in P?

A. Given a Boolean formula, is it satisfiable?

B. Given a graph and two vertices, is there a path between them?

C. Given a graph, does it have a clique of size k?

D. Given an integer n, is it the product of two primes?

Answer: B

10. If A <=p B and B in P, then:

A. A in NP

B. A in P

C. B is NP-complete

D. A = B

Answer: B

11. Let A be a language decided by a deterministic polynomial-time algorithm. What can we conclude?

A. A in NP and A in coNP

B. A not in NP but A in P

C. A in NP, but it is undecidable

D. A is not closed under reduction

Answer: A

12. Which of the following implications always holds?

A. A in NP -> A in P

B. A in coNP -> A in NP

C. A in P -> A in NP

D. A in NP -> not A in NP

Answer: C

13. A verifier for a language A runs in polynomial time and receives input x and certificate y. What must be true for A in NP?

A. The certificate must always be of constant size

B. For every x in A, there exists a certificate y of polynomial length such that the verifier accepts (x, y)

C. The verifier must reject all inputs x not in A for every certificate

D. The verifier must explore all possible certificates in polynomial time

Answer: B

14. Which of the following problems is in P?

A. Given a Boolean formula, is it satisfiable?

B. Given a graph and two vertices, is there a path between them?

C. Given a graph, does it have a clique of size k?

D. Given an integer n, is it the product of two primes?

Answer: B

15. If A <=p B and B in P, then:

A. A in NP

B. A in P

C. B is NP-complete

D. A = B

Answer: B

16. Let A be a language decided by a deterministic polynomial-time algorithm. What can we conclude?

A. A in NP and A in coNP

B. A not in NP but A in P

C. A in NP, but it is undecidable

D. A is not closed under reduction

Answer: A

17. Which of the following implications always holds?

A. A in NP -> A in P

B. A in coNP -> A in NP

C. A in P -> A in NP

D. A in NP -> not A in NP

Answer: C

18. A verifier for a language A runs in polynomial time and receives input x and certificate y. What must be true for A in NP?

A. The certificate must always be of constant size

B. For every x in A, there exists a certificate y of polynomial length such that the verifier accepts (x, y)

C. The verifier must reject all inputs x not in A for every certificate

D. The verifier must explore all possible certificates in polynomial time

Answer: B

19. Which of the following problems is in P?

A. Given a Boolean formula, is it satisfiable?

B. Given a graph and two vertices, is there a path between them?

C. Given a graph, does it have a clique of size k?

D. Given an integer n, is it the product of two primes?

Answer: B

20. If A <=p B and B in P, then:

A. A in NP

B. A in P

C. B is NP-complete

D. A = B

Answer: B

21. Let A be a language decided by a deterministic polynomial-time algorithm. What can we conclude?

A. A in NP and A in coNP

B. A not in NP but A in P

C. A in NP, but it is undecidable

D. A is not closed under reduction

Answer: A

22. Which of the following implications always holds?

A. A in NP -> A in P

B. A in coNP -> A in NP

C. A in P -> A in NP

D. A in NP -> not A in NP

Answer: C

23. A verifier for a language A runs in polynomial time and receives input x and certificate y. What must be true for A in NP?

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B. For every x in A, there exists a certificate y of polynomial length such that the verifier accepts (x, y)

C. The verifier must reject all inputs x not in A for every certificate

D. The verifier must explore all possible certificates in polynomial time

Answer: B

24. Which of the following problems is in P?

A. Given a Boolean formula, is it satisfiable?

B. Given a graph and two vertices, is there a path between them?

C. Given a graph, does it have a clique of size k?

D. Given an integer n, is it the product of two primes?

Answer: B

25. If A <=p B and B in P, then:

A. A in NP

B. A in P

C. B is NP-complete

D. A = B

Answer: B

26. Let A be a language decided by a deterministic polynomial-time algorithm. What can we conclude?

A. A in NP and A in coNP

B. A not in NP but A in P

C. A in NP, but it is undecidable

D. A is not closed under reduction

Answer: A

27. Which of the following implications always holds?

A. A in NP -> A in P

B. A in coNP -> A in NP

C. A in P -> A in NP

D. A in NP -> not A in NP

Answer: C

28. A verifier for a language A runs in polynomial time and receives input x and certificate y. What must be true for A in NP?

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C. The verifier must reject all inputs x not in A for every certificate

D. The verifier must explore all possible certificates in polynomial time

Answer: B

29. Which of the following problems is in P?

A. Given a Boolean formula, is it satisfiable?

B. Given a graph and two vertices, is there a path between them?

C. Given a graph, does it have a clique of size k?

D. Given an integer n, is it the product of two primes?

Answer: B

30. If A <=p B and B in P, then:

A. A in NP

B. A in P

C. B is NP-complete

D. A = B

Answer: B

31. Let A be a language decided by a deterministic polynomial-time algorithm. What can we conclude?

A. A in NP and A in coNP

B. A not in NP but A in P

C. A in NP, but it is undecidable

D. A is not closed under reduction

Answer: A

32. Which of the following implications always holds?

A. A in NP -> A in P

B. A in coNP -> A in NP

C. A in P -> A in NP

D. A in NP -> not A in NP

Answer: C

33. A verifier for a language A runs in polynomial time and receives input x and certificate y. What must be true for A in NP?

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B. For every x in A, there exists a certificate y of polynomial length such that the verifier accepts (x, y)

C. The verifier must reject all inputs x not in A for every certificate

D. The verifier must explore all possible certificates in polynomial time

Answer: B

34. Which of the following problems is in P?

A. Given a Boolean formula, is it satisfiable?

B. Given a graph and two vertices, is there a path between them?

C. Given a graph, does it have a clique of size k?

D. Given an integer n, is it the product of two primes?

Answer: B

35. If A <=p B and B in P, then:

A. A in NP

B. A in P

C. B is NP-complete

D. A = B

Answer: B

36. Let A be a language decided by a deterministic polynomial-time algorithm. What can we conclude?

A. A in NP and A in coNP

B. A not in NP but A in P

C. A in NP, but it is undecidable

D. A is not closed under reduction

Answer: A

37. Which of the following implications always holds?

A. A in NP -> A in P

B. A in coNP -> A in NP

C. A in P -> A in NP

D. A in NP -> not A in NP

Answer: C

38. A verifier for a language A runs in polynomial time and receives input x and certificate y. What must be true for A in NP?

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C. The verifier must reject all inputs x not in A for every certificate

D. The verifier must explore all possible certificates in polynomial time

Answer: B

39. Which of the following problems is in P?

A. Given a Boolean formula, is it satisfiable?

B. Given a graph and two vertices, is there a path between them?

C. Given a graph, does it have a clique of size k?

D. Given an integer n, is it the product of two primes?

Answer: B

40. If A <=p B and B in P, then:

A. A in NP

B. A in P

C. B is NP-complete

D. A = B

Answer: B

41. Let A be a language decided by a deterministic polynomial-time algorithm. What can we conclude?

A. A in NP and A in coNP

B. A not in NP but A in P

C. A in NP, but it is undecidable

D. A is not closed under reduction

Answer: A

42. Which of the following implications always holds?

A. A in NP -> A in P

B. A in coNP -> A in NP

C. A in P -> A in NP

D. A in NP -> not A in NP

Answer: C

43. A verifier for a language A runs in polynomial time and receives input x and certificate y. What must be true for A in NP?

A. The certificate must always be of constant size

B. For every x in A, there exists a certificate y of polynomial length such that the verifier accepts (x, y)

C. The verifier must reject all inputs x not in A for every certificate

D. The verifier must explore all possible certificates in polynomial time

Answer: B

44. Which of the following problems is in P?

A. Given a Boolean formula, is it satisfiable?

B. Given a graph and two vertices, is there a path between them?

C. Given a graph, does it have a clique of size k?

D. Given an integer n, is it the product of two primes?

Answer: B

45. If A <=p B and B in P, then:

A. A in NP

B. A in P

C. B is NP-complete

D. A = B

Answer: B

46. Let A be a language decided by a deterministic polynomial-time algorithm. What can we conclude?

A. A in NP and A in coNP

B. A not in NP but A in P

C. A in NP, but it is undecidable

D. A is not closed under reduction

Answer: A

47. Which of the following implications always holds?

A. A in NP -> A in P

B. A in coNP -> A in NP

C. A in P -> A in NP

D. A in NP -> not A in NP

Answer: C

48. A verifier for a language A runs in polynomial time and receives input x and certificate y. What

must be true for A in NP?

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B. For every x in A, there exists a certificate y of polynomial length such that the verifier accepts (x, y)

C. The verifier must reject all inputs x not in A for every certificate

D. The verifier must explore all possible certificates in polynomial time

Answer: B

49. Which of the following problems is in P?

A. Given a Boolean formula, is it satisfiable?

B. Given a graph and two vertices, is there a path between them?

C. Given a graph, does it have a clique of size k?

D. Given an integer n, is it the product of two primes?

Answer: B

50. If A <=p B and B in P, then:

A. A in NP

B. A in P

C. B is NP-complete

D. A = B

Answer: B